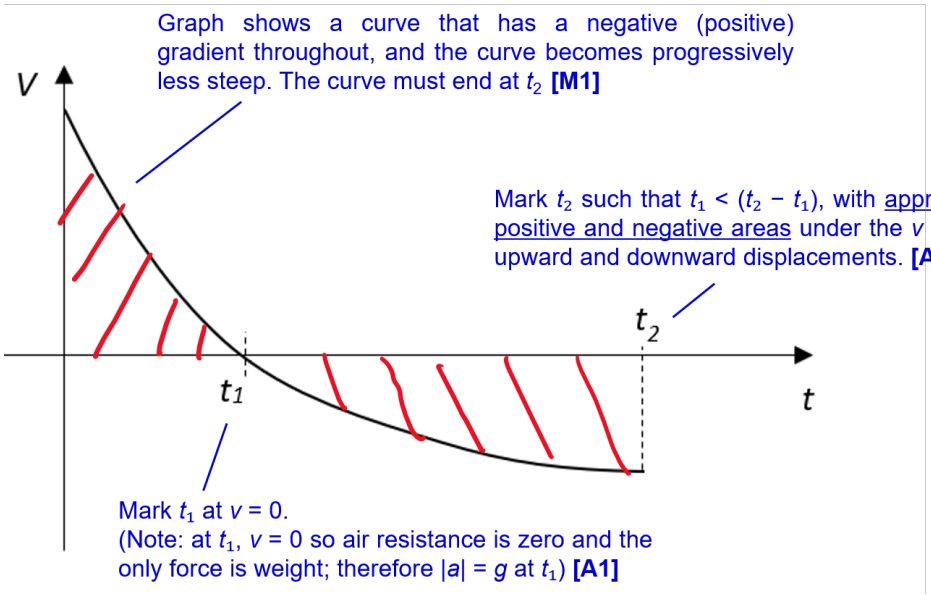
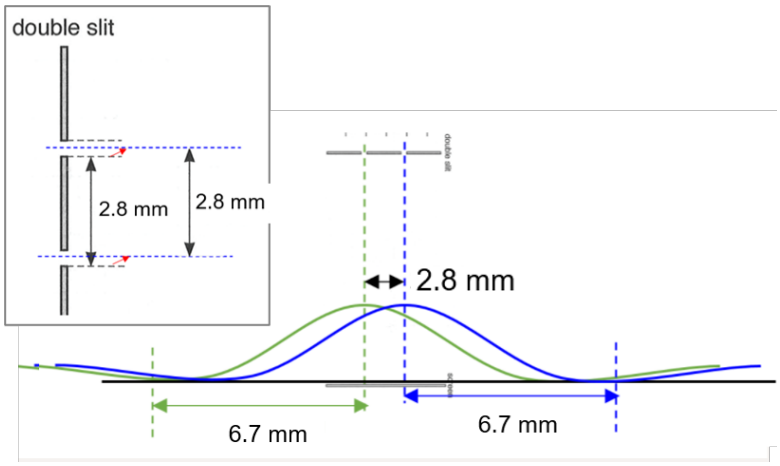
2026 EJC H2 Physics P2 Solutions

Qns	Answer	Marks
1(a)(1)	At t = 0, the gradient of the graph is 0. Hence the speed of the ball is 0.	A1
1(a)(2)	The gradient of the graph eventually becomes constant after the t = 0.80 s mark, indicating that the ball has reached terminal velocity. The acceleration, and thus the resultant force is zero. Since $F_{net} = mg - F_{air} = 0$, this shows that air resistance is significant and balances the weight when terminal velocity is reached.	M1 A1
1(b)	Draw a tangent to Fig. 1.1 at t = 0.40 s and determine its gradient. Speed = (2.8 ± 0.3) m s^{-1}.	M1 A1
1(c)		M1 A1 A1
	TOTAL FOR QUESTION 1	**8**
2(a)	The rate of change of momentum of a body is (directly) proportional to the resultant force acting on the body and is in the same direction as the resultant force.	B1
2(b)(i)	The sand initially has zero **horizontal** momentum but accelerates to the speed of the belt. Therefore, the belt exerts a forward horizontal force on the sand to produce this change in momentum. By Newton's third law, the sand exerts an equal and opposite horizontal force on the belt.	M1 A1
2(b)(ii)	$\langle F \rangle = \frac{m}{t}\Delta v = (0.100)(0.050)$ $= 0.0050$ N	M1 A0
2(b)(iii)1	$P = Fv = 0.0050 \times 0.050 = 2.5 \times 10^{-4}$ W	B1
2(b)(iii)2	In one second, rate at which sand gains kinetic energy, K:	

Qns	Answer	Marks
	$K = \frac{\frac{1}{2}m\Delta(v^2)}{t} = \frac{1}{2}\left(\frac{m}{t}\right)\left(v_f^{\,2} - v_i^{\,2}\right)$ $= \frac{1}{2}\left(\frac{100}{1000}\right)\left(\left(\frac{5.0}{100}\right)^2 - 0\right)$ $= 1.25 \times 10^{-4}\,\text{W}$	**B1**
2(b)(iv)	Perfectly inelastic collision. Energy transferred mechanically via work done by the motor is to the (horizontal)l kinetic store of the sand and to the internal store of the system (heating/friction).	**B1** **B1**
	TOTAL FOR QUESTION 2	**8**
3(a)	Electric potential at a point is the work done per unit charge by an external force in bringing a small positive test charge from infinity to that point.	**B1**
3(b)(i)	$E = -\frac{dV}{dr}$ $\lvert E \rvert$ = gradient of graph $= \left\lvert \frac{2.50 - 0}{(0 - 2.50) \times 10^{-2}} \right\rvert$ [M1] $= 100\,\text{N C}^{-1}$ [A1] Acceptable range: 88 <=E<= 1105□□ Direction: in positive x-direction / towards right / from left to right / → [B1]	**M1** **A1** **B1**
3(b)(ii)	The graph is symmetrical.	**B1**
3(b)(iii)	Using a point such as x = 0.20 cm where V ≈ 6.0 V: $V = \frac{Q}{4\pi\varepsilon_o x} + \frac{Q}{4\pi\varepsilon_o (0.040 - x)}$ $6.0 = \frac{Q}{4\pi\varepsilon_o}\left(\frac{1}{0.002} + \frac{1}{0.040 - 0.002}\right)$ [M1] $Q = 1.27 \times 10^{-12}\,\text{C}$ [A1] Accept also if student used other suitable points for substitution.	**M1** **A1**
	TOTAL FOR QUESTION 3	**7**
4(a)	Resistance of a circuit component is the ratio of the potential difference across the component to the current passing through it.	**B1**

Qns	Answer	Marks
4(b)	e.m.f. is the energy converted from non-electrical to electrical form per unit charge, while p.d. is the energy converted from electrical to non-electrical form per unit charge. [B1]	**B1**
4(c)	As R_2 increases, the total resistance of the circuit increases and the current in the cell decreases. The p.d. across the internal resistance therefore decreases. Hence, using $V_{terminal} = E - Ir$, the terminal p.d. increases. Since R_1 is constant, the current through R_1 increases. -	**M1** **M1** **A1**
4(d)(i)	p.d. across r = 1.40 − (0.220 × 5.50) = 0.19 V. Icell = 0.19/0.13 = 1.46 A ≈ 1.5 A (2 s.f.).	**C1** **A1**
4(d)(ii)	I_2 = 1.5 − 0.220 = 1.28 A; R_2 = (0.220×5.50)/1.28 = 0.945 Ω	**C1** **A1**
	TOTAL FOR QUESTION 4	**9**
5(a)	For a single-slit diffraction pattern, $\sin\theta = \frac{\lambda}{b} \Rightarrow \theta = \sin^{-1}\left(\frac{\lambda}{b}\right) = \sin^{-1}\left(\frac{670\times10^{-9}}{0.12\times10^{-3}}\right) = 0.00558336 \text{ rad (6.s.f)}$ Let the distance between the central maxima to first minima for the diffraction pattern be *y*. Hence, $\tan\theta = \frac{y}{D} \Rightarrow y = D\ (\tan\theta)$ $= 1.20\ (\tan(0.00558336))$ $= 0.00670010 \text{ m (6.s.f)}$ Thus, the width of the central fringe is: 2*y* = 2 (0.00670010) = 0.0134 m (3.s.f) = 13.4 mm (shown) **Note:** Deduct 1 mark if students - used small angle approximation without justification - write the angle in degrees without indicating the degree symbol	**C1** **C1** **M1** **A0**
5(b)(i)	The distance from the central bright maximum to first minimum is: 13.4 / 2 = 6.7 mm. However, the central bright maxima are 2.8 mm away from each other. Hence, we cannot distinguish between the two diffraction patterns from each single slit / they cannot be resolved.	**M1** **A1**

Qns	Answer	Marks
		
5(b)(ii)	$x = \frac{D\lambda}{a} = \frac{(1.2)(670\times10^{-9})}{(2.8\times10^{-3})} = 0.000287143 \text{ m (6.s.f)}$ An estimation of the number of fringes, *n* is $n = \frac{0.0134002}{0.000287143} = 46.66733 = 46$ (round down)	**C1** **A1**
	TOTAL FOR QUESTION 5	**7**
6(a)	Faraday's law of electromagnetism states that the (induced) e.m.f. is directly proportional to the rate of change of magnetic flux linkage.	**B1**
6(b)(i)	$\varepsilon = \frac{d\phi}{dt} = \frac{\Delta\phi}{\Delta t}$ $\Delta\phi = \varepsilon \times \Delta t$ $= 0.55 \times 14 = 7.7$ Wb Accepted units include: Wb / Vs / Tm^2	**A1** **A1**
6(b)(ii)	$\phi = BA$ $A = \frac{\phi}{B} = \frac{7.7}{38\times10^{-6}}$ $= 2.0\times10^{5}$ m^2 **Note:** Allow ECF from **(b)(i)**	**A1**
6(b)(iii)	**Method 1:** Area = speed x time x width $v = \frac{A}{t \times \text{width}}$ $= \frac{2.0\times10^{5}}{14\times68}$ $= 210$ m s^{-1}	**C1** **A1**

Qns	Answer	Marks
	OR **<u>Method 2:</u>** $\varepsilon = \frac{d\phi}{dt}$ $\varepsilon = BLv$ } **C1** $0.55 = \left(38 \times 10^{-6}\right)(68)v$ $v = 213 \text{ m s}^{-1}$ $= 210 \text{ m s}^{-1}$ (2 s.f.) **<u>Note:</u>** Allow ECF from **(b)(ii)**	**OR** **C1** **A1**
6(b)(iv)	As the aircraft travels through the magnetic field with a velocity perpendicular to the field lines, charges in the wing experience a magnetic force. Using Fleming's left-hand rule, magnetic forces act on positive (negative) charges, causing them to move toward wingtip P (Q). As positive (negative) charges accumulate at P (Q) wingtip P will be at a higher potential than Q. sp	**B1** **A1**
6(c)(i)	<table><tr><th>Resistor</th><th>r.m.s. voltage, $V_{r.m.s.}$ / V</th><th>peak voltage, V_0 / V</th></tr><tr><td>**R_1**</td><td>$\frac{16}{\sqrt{2}}$ = 11 (or 11.3)</td><td>16</td></tr><tr><td>**R_2**</td><td>$\frac{16}{2}$ = 8.0</td><td>16</td></tr></table>	**A1**
6(c)(ii)	The peak voltages are the same because when the diode is forward-biased, its <u>resistance is close to zero</u>, giving both resistors the same peak potential difference. The r.m.s. voltages for both resistors are different because the branch with resistor R_2 and the diode undergoes <u>half-wave rectification</u>, unlike the branch with only resistor R_1.	**B1** **B1**
	TOTAL FOR QUESTION 6	**11**
7(a)	Half-life is the <u>average</u> time taken for half the original <u>number</u> of radioactive <u>nuclei</u> to decay.	**B1**
7(b)(i)	$A = \lambda N$ $1.6 \times 10^{14} = \frac{ln\,2}{3.6 \times 24 \times 3600} N$ [M1] $N = 7.18 \times 10^{19}$ Mass = $7.18 \times 10^{19} \times 224 \times 1.66 \times 10^{-27}$ [M1] $= 2.67 \times 10^{-5}$ kg $= 27$ mg [A0]	**M1** **M1** **A0**

Qns	Answer	Marks
7(b)(ii)	Consider activity of 5.0mg of the fresh sample $A_o = \frac{5.0}{2.67} \times 1.6 \times 10^{14}$ $= 2.996 \times 10^{13}$ Bq	M1
	$\frac{A}{A_o} = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$ $\frac{2.1 \times 10^{10}}{2.996 \times 10^{13}} = \left(\frac{1}{2}\right)^{\frac{t}{3.6}}$	M1
	$t = 38$ days	A1
	TOTAL FOR QUESTION 7	**6**

8(a)(i)	$T = RC\ln\left(\frac{\varepsilon - V_s}{\varepsilon - V_b}\right)$ $= 18\times10^{3}\times33\times10^{-6}\ln\left(\frac{5.4-0.9}{5.4-4.5}\right)$ [C1] $= 0.956\text{s}$ $f = \frac{1}{T} = 1.046\,\text{s}^{-1}$ = 1.05 Hz [M1] Comment: Passage mentioned: steady heart rate is 60 beats per minute, the frequency of 1.046 s^{-1} is about 62 beats per min which is close to steady heart rate. Hence it is appropriate. [A1] *Candidate must compare current frequency with steady heart rate from passage to gain this mark.* Award [B1] for any one physiologically valid situation where a fixed pacing rate is inappropriate, with a brief explanation of why. Examples: **Physical activity**: cannot increase rate during exercise leading to insufficient cardiac output during exertion **Sleep**: healthy hearts slow to 40–50 bpm during deep sleep hence fixed rate may be unnecessarily high, consuming more battery **Emotional stress/anxiety**: heart rate naturally rises and fixed rate cannot respond **Illness/fever**: body may require higher heart rate and fixed rate cannot adapt **Individual variation**: 60 bpm may not be the optimal resting rate for every patient but there is no way to adjust without surgery to replace resistor **Competing with intrinsic rhythm**: if patient's own heart occasionally fires, fixed-rate pacemaker may deliver a pulse at the wrong moment in the cardiac cycle, potentially dangerous p.s. this is actually the most clinically serious one, known as "R-on-T phenomenon"	**C1** **M1** **A1** **B1**
8(a)(ii)	τ_charging = RC = $18\times10^{3}\times33\times10^{-6}$ = 0.594 s □ τ_discharge = R_L C = $500\times33\times10^{-6}$ = 0.0165 s [M1 for both calculations] □ τ_discharge is 36× smaller than τ_charging (or compare and say about 2% only) ; since discharge time << charging time << T, the assumption that discharge is almost instantaneous is well justified. [A1]	**M1** **A1**
8(b)(i)	The energy *U* stored in a capacitor of capacitance *C* charged to potential difference *V* is: $U = \frac{1}{2}CV^2$ Since the capacitor charges to V_b before each discharge and retains charge at *Vs* after each discharge, the energy ΔU delivered to the heart per pulse is: $\Delta U = \frac{1}{2}C\left(V_b^{\,2} - V_s^{\,2}\right)$ = ½ (33 x 10^{-6})($4.5^2 - 0.9^2$) = 3.21 x 10^{-4} J	**C1** **M1** for correct working + substitution A0
8(b)(ii)	Power = energy / time = energy x frequency	

	P = 3.21 x 10^{-4} x 1.046 = 0.0003357 = 3.36 x 10^{-4} W	M1 A1
8(b)(iii)	Each pulse is said to be almost "instantaneous" which means the power is delivered only during the discharging process [B1] and during the charging process, no energy is delivered to the heart tissue [B1]; hence the time-averaged power is much lower than the power during each pulse	B1 B1
8(c)(i)	Converts capacity to coulombs: Q = 1.2 × 3600 = 4320 C **[C1]** Energy = VQ = 5.4 × 4320 = 23328 J = 23.3 kJ (3sf) **[A1]**	C1 A1
8(c)(ii)	Lifespan = E_batt / P_avg = 23328 / (3.36×10^{-4}) = 6.94×10^{7} s = 2.2 years Very short lifespan, will require operation every 2.2 years to change battery. OR 2.2 years is an overestimate because the calculation assumes all stored battery energy is ultimately delivered to the heart and neglects losses/other circuit consumption.	M1 A1
8(d)(i)	Increase in C → increase in T calculated from (a)(i)/ time constant → longer charging time and longer period T → lower pacing frequency since f = 1/T. [1]	B1
8(d)(ii)	A patient with complete heart block is entirely dependent on the pacemaker for ventricular stimulation; if the pacing rate drops, their heart rate drops correspondingly. [B1] If the pacing rate falls below ~40 bpm (as stated in passage), the patient risks fainting / cardiac arrest. [B1] A slow gradual drift may go unnoticed without regular monitoring, person's health may deteriorate slowly, making it potentially life-threatening / dangerous. [B1] Award max 2 points for any 2 points.	B1 B1
	TOTAL FOR QUESTION 8	**19**